Aliter

We find that direction ratios of the straight line joining the points $A(6,7,5)$ and $B(8,10,6)$ are $(b_1,b_2,b_3)=(2,3,1)$ and direction ratios of the line joining the points $C(10,2,-5)$ and $D(8,3,-4)$ are $(d_1,d_2,d_3)=(-2,1,1)$. Since $b_1d_1+b_2d_2+b_3d_3=(2)(-2)+(3)(1)+(1)(1)=0$, the two straight lines are perpendicular. ■

Example 6.32

Show that the lines $\frac{x-1}{4}=\frac{2-y}{6}=\frac{z-4}{12}$ and $\frac{x-3}{-2}=\frac{y-3}{3}=\frac{5-z}{6}$ are parallel.

Solution

We observe that the straight line $\frac{x-1}{4}=\frac{2-y}{6}=\frac{z-4}{12}$ is parallel to the vector $4\hat{i}-6\hat{j}+12\hat{k}$ and the straight line $\frac{x-3}{-2}=\frac{y-3}{3}=\frac{5-z}{6}$ is parallel to the vector $-2\hat{i}+3\hat{j}-6\hat{k}$.

Since $4\hat{i}-6\hat{j}+12\hat{k}=-2(-2\hat{i}+3\hat{j}-6\hat{k})$, the two vectors are parallel, and hence the two straight lines are parallel. ■

EXERCISE 6.4

1. Find the non-parametric form of vector equation and Cartesian equations of the straight line passing through the point with position vector $4\hat{i}+3\hat{j}-7\hat{k}$ and parallel to the vector $2\hat{i}-6\hat{j}+7\hat{k}$.
2. Find the parametric form of vector equation and Cartesian equations of the straight line passing through the point $(-2,3,4)$ and parallel to the straight line $\frac{x-1}{-4}=\frac{y+3}{5}=\frac{8-z}{6}$.
3. Find the points where the straight line passes through $(6,7,4)$ and $(8,4,9)$ cuts the xz and yz planes.
4. Find the direction cosines of the straight line passing through the points $(5,6,7)$ and $(7,9,13)$. Also, find the parametric form of vector equation and Cartesian equations of the straight line passing through two given points.
5. Find the acute angle between the following lines.
   (i) $\vec{r}=(4\hat{i}-\hat{j})+t(\hat{i}+2\hat{j}-2\hat{k}),\ \vec{r}=(\hat{i}-2\hat{j}+4\hat{k})+s(-\hat{i}-2\hat{j}+2\hat{k})$
   (ii) $\frac{x+4}{3}=\frac{y-7}{4}=\frac{z+5}{5},\ \vec{r}=4\hat{k}+t(2\hat{i}+\hat{j}+\hat{k})$.
   (iii) $2x=3y=-z$ and $6x=-y=-4z$.
6. The vertices of ΔABC are $A(7,2,1), B(6,0,3)$, and $C(4,2,4)$. Find $\angle ABC$.
7. If the straight line joining the points $(2,1,4)$ and $(a-1,4,-1)$ is parallel to the line joining the points $(0,2,b-1)$ and $(5,3,-2)$, find the values of a and b.
8. If the straight lines $\frac{x-5}{5m+2}=\frac{2-y}{5}=\frac{1-z}{-1}$ and $x=\frac{2y+1}{4m}=\frac{1-z}{-3}$ are perpendicular to each other, find the value of m.
9. Show that the points $(2,3,4),(-1,4,5)$ and $(8,1,2)$ are collinear.

12th_Maths_EM_Vol1_CH 6 Vector Algebra.indd 249 16-12-2022 14:25:21